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1. Many of the inherent safety features of SFRs derive from the sodium coolant. Na has a relatively low melting temperature and a high boiling point, which allows for operation near atmospheric pressure (no pressurization) and has a high margin to boiling which is beneficial in accident scenarios. The Na also has a long thermal response time, which reduces the impacts of transients. Na is not corrosive on its own, although the environment can lead to corrosion.

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-negative reactivity feedback

2. There are several reasons to alloy uranium-metal fuels. Originally, alloying was used to reduce the severe swelling of the anisotropic orthorhombic α-phase by acting as point-defect sinks (U-fissium fuels). Higher-content alloys serve to stabilize the high-temperature BCC γ-phase at lower temperatures by lowering the phase boundary - this also has the additional impact of raising T_{melt}. The best alloying elements for this purpose are the transition metals Zr & Mo, with the latter used at 10 wt.% in fast reactor applications and the latter at 7-12 wt% in thermal-spectra research & test reactors. In addition to phase-stabilization and swelling mitigation, Zr can also serve to slow diffusion rates (reducing FCCI) and can form a protective rind.

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3. Metallic fuels suffer from very large, rapid swelling (30% by 5% BU). In early designs, the fuel swelling would result in severe FCMI, which lead to rupture and limited the reactors to low BU. At about 30% swelling, the majority of the fission gas escapes the fuel, leaving only the less severe solid-F.P. swelling (~1.5% per %BU). The low smear density allows the fuel to swell to the point of gas release before coming into contact with the cladding. The large plenum collects the released gas, and its size (1-1.5x the fuel slug length) reduces the fission gas pressure and reduces strain on the cladding.

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4. Constituent redistribution is primarily the movement of Zr within the fuel. This occurs for largely two reasons: chemical concentration gradients and Soret diffusion. At the typical operating temperature of metallic fuels, there are ~6 different U-Pu-Zr phases, each with different solubilities of Zr & Pu. The center of the fuel is the hottest and is dominated by the γ-phase (high Zr solubility) and the colder periphery is a mix of the low-temperature low-solubility α-phase and the intermediate higher-solubility δ-phase. Zr and U respond differently to the thermal gradient, where U moves down (Q^* ~ 5) and Zr moves up (Q^* ~ -34). This leads to 3 zones, from center-to-periphery: the Zr-enriched zone (γ-UZr), the Zr-depleted zone (β-UZr), and the as-sintered region (α-UZr + δ-UZr eutectoid). The different phases have different porosity morphologies - large interconnected bubbles in the γ-phase and smaller pores filling the lamnar α/δ phase. This results in anisotropic swelling of the fuel slug as a whole (larger radially than axially).

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5. The thermal conductivity of metallic fuels intitially decreases with increasing swelling, since the primary degredation of k is porosity. This trend continues until the fission gas is released at ~30% swelling, at which point the bond sodium infiltrates the porosity. This results in a sharp increase of the fuel's overall k. Since the Na which occupies ~15-20% of the porosity has a very high k value. At this point, k remains relatively constant.

6. In metallic fuel, FCCI takes the form of interdiffusion between the fuel & cladding species. In addition to weakening the mechanical properties of the cladding, low-melt phases can form, primarily (U/Pu)$_6$Fe and (U/Pu)Fe$_2$, which have a low eutectic melting temperature of ~725°C. The speed of the diffusion can be limited by Zr, and a protective (likely ZrO_2) rind can form if there are enough impurities (O, C, N), which limits FCCI. Zr also mitigates FCCI by forming phases w/ Fe which have a higher Tmelt than the U/Pu-Fe phases. The cladding elements which participate are primarily Fe and Cr, but Ni as well. The primary fuel species are the lanthanides - while U and Pu also participate to a certain extent, the lanthanides (particularly Nd) penetrate furthest into the cladding.

Ce & Nd

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7. MOX fuels have distinct zones due to the steep temperature gradients resulting from the low thermal conductivity of MOX: ① the central void, ② the columnar grain region, ③ the equiaxed region, and ④ the as-sintered region. The central void forms quickly due to the very high linear heat rate of the MOX fuel and the high fuel centerline temperature. The voids travel up the thermal gradients, concentrating in the center of the fuel. The voids migrate via an evaporation-condensation mechanism: the thermal gradients within the void is higher than that of the fuel, so matter evaporated from the hot side and is deposited on the cold side in a single crystal fashion. The elongated lenticular voids destroy the microstructure as they rise, creating long columnar grains. In the cold region of the fuel, temperatures are high enough to result in grain growth but not high enough to cause the vaporization migration mechanism - this is the equiaxed grain region. At the coldest part of the fuel, the periphery, temperatures are not high enough to cause any significant alteration to the microstructure.

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8. Pu varies in 2 ways. Preferential evaporation of UO_2 during the pore migration (UO_2 has a higher vapor pressure) leads to Pu (and fissile) enrichment and U depletion around the central void. Pu agglomerates can also be present throughout the fuel due to poor mixing during fabrication. Oxygen migrates down the temperature gradients to the edge of the fuel, leading to stoichiometric O/M at the edge & lower O/M ratios towards the center. Oxygen potential increases until Mo oxidizes, and then stabilizes at stoichiometric levels.

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9. Joint Oxide Gain (JOG) is a Cs-Mo oxide which condenses on the cladside of the fuel pin. It forms because the volatile F.P.s (Cs, Te, I) and O migrate to the edge of the fuel. Mo becomes volatile in the presence of both Cs and O. These volatiles evaporate, move down the temperature gradients, and condense. JOG enhances thermal conductivity, serves as a soft mechanical buffer, and reduces swelling. However, it also facilitates FCCI, which includes corrosion by primarily the volatile species Cs, Te, and I. Cs specifically corrodes the protective chromia layer of the stainless steel cladding in the presence of excess oxygen.

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10. Sodium corrosion occurs via 2 mechanisms: dissolution of structural components in Na and chemical reactions with impurities in Na. Solubility increases with temp. and is particularly high for Ni - this can lead to thermal gradient mass transfer, where soluble species are moved from hot to cold regions. It can also lead to (de)carburization via dissolution of C (mitigated by higher Cr contents). The presence of impurities, particularly oxygen, increases the effective solubility of Fe & Cr via chemical reactions with Na_2O. Corrosion rate is generally dependent on temperature, coolant velocity, dissolved oxygen contents, immersion time, and (de)carburization. Immersion time is more impactful for austenitic steels, where the Ni dissolution dominates before steady-state Fe dissolution.